PERCENTAGES

A useful method of simplifying calculations involving percentages is to reduce one difficult percentage into smaller and easier-to-calculate parts. In the example below, the smaller percentages include 10% and 5%, which are easy to work out.

▷ **Adding 17.5 percent**
Here a shop wants to charge \$480 for a new bike. However, the owner of the shop has to add a sales tax of 17.5 percent to the price. How much will it then cost?

sales tax

17.5% of 480

original price of bike

10% of 480 = 48

5% of 480 = 24

2.5% of 480 = 12

2.5% of 480 is half of 5% of 480, which is half of 10% of 480

$$\begin{array}{r} 48 \\ 24 \\ +\ 12 \\ \hline 84 \\ \hline \end{array}$$

results added together

84 is 17.5% of 480

First, write down the percentage price increase required and the original price of the bike.

Next, reduce 17.5% into the easier stages of 10%, 5%, and 2.5% of \$480, and calculate their values.

The sum of 48, 24, and 12 is 84, so \$84 is added to \$480 for a price of \$564.

Switching

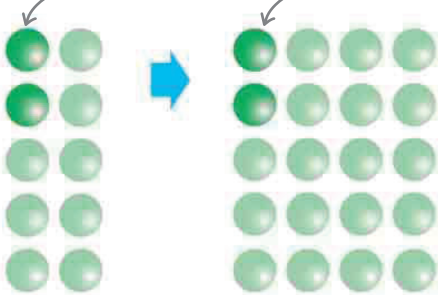
A percentage and an amount can both be "switched," to produce the same result with each switch. For example, 50% of 10, which is 5, is exactly the same as 10% of 50, which is 5 again.

20% is 2 of 10 balls | amount of balls is 10 | 10% is 2 of 20 balls | amount of balls is 20

20% of 10 = 10% of 20

2 balls are 20% of 10 balls

2 balls are 10% of 20 balls



Progression

A progression involves dividing the percentage by a number and then multiplying the result by the same number. For example, 40% of 10 is 4. Dividing this 40% by 2 and multiplying 10 by 2 results in 20% of 20, which is also 4.

40% is 4 of 10 balls | amount of balls is 10 | 20% is 4 of 20 balls | amount of balls is 20

40% of 10 = 20% of 20

4 balls are 40% of 10 balls

4 balls are 20% of 20 balls